Semirings for database provenance

Yiannis Hadjicharalambous

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What is a database?

Loans

Loan_id	Member_id	Book_id	Borrow_date	Duration
43666	4221	389	17-02-2026	7
43666	4221	450	17-02-2026	7
43952	4222	975	17-02-2026	30
54673	4223	389	06-03-2026	14
58930	4224	975	09-03-2026	30
58930	4224	879	09-03-2026	30

Books

Book_id	Book_title	Genre	Copies
389	Nineteen Eighty-Four	Fiction	5
450	Brave New World	Fiction	4
879	Divine Comedy	Poetry	6
975	Critique of Pure Reason	Philosophy	3

Members

Member_id	First_name	Last_name
4221	Aditya	Prakash
4222	Graham	Leigh
4223	Ivan	Di Liberti
4224	Luigi	Brunetta

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$\{(\textbf{Member\_id}, 4221), (\textbf{First\_name}, \text{Aditya}), (\textbf{Last\_name}, \text{Prakash})\} \in$ *Members*

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The *relational algebra* consists of the following operations:

- Selection: $\sigma_\theta(R)$
- Projection: $\pi_L(R)$
- Renaming: $\rho_\beta(R)$
- Natural join: $R \bowtie S$
- Union: $R \cup S$
- Difference: $R - S$

Example query

Query: Find the title of every book borrowed for 7 or 14 days.

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$$\pi_{Book\_title}\Big(\sigma_{Duration=7}(Loans \bowtie Books) \cup \sigma_{Duration=14}(Loans \bowtie Books)\Big)$$

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We represent it by tagging tuples with annotations.

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54673	4223	389	06-03-2026	14	l_4
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- An *S-relation* as a function $R : \mathbb{D}^U \to S$ with finite support.
- An *S-database* as a finite set of S-relations.

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We obtain $(S, +, \cdot, 0, 1)$.

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Book_title	
Nineteen Eighty-Four	$l_1 \cdot b_1 + l_4 \cdot b_1$
Brave New World	$l_2 \cdot b_2$

When do semirings come in?

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For all relations R, S, T over attribute set U:

$$
\begin{aligned}
R \cup (S \cup T) &= (R \cup S) \cup T \\
R \cup S &= S \cup R \\
R \cup \mathbf{0}_U &= R \\
R \bowtie (S \bowtie T) &= (R \bowtie S) \bowtie T \\
R \bowtie S &= S \bowtie R \\
R \bowtie \mathbf{1} &= R \\
R \bowtie (S \cup T) &= (R \bowtie S) \cup (R \bowtie T) \\
R \bowtie \mathbf{0}_U &= \mathbf{0}_U
\end{aligned}
$$

What is a (commutative) semiring?

An algebraic structure $(S, +, \cdot, 0, 1)$ *such that for all* $a, b, c \in S$*:*

$$
\begin{gathered}
a + (b + c) = (a + b) + c \\
a + b = b + a \\
a + 0 = a \\
a \cdot (b \cdot c) = (a \cdot b) \cdot c \\
a \cdot b = b \cdot a \\
a \cdot 1 = a \\
a \cdot (b + c) = (a \cdot b) + (a \cdot c) \\
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\end{gathered}
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\end{aligned}
$$

Examples:

- $\mathbb{N} = (\{0, 1, 2, \dots\}, +, \times, 0, 1)$
- $\mathbb{B} = (\{0, 1\}, \vee, \wedge, \bot, \top)$

Extending the framework

Further research has focused on extending the framework beyond the positive relational algebra to more expressive queries.

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We briefly consider one case, aggregation.

Aggregate queries

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title	genre	price
Book 1	biography	40
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Book 3	poetry	60
Book 4	biography	30
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$\gamma_{genre, AVG(price)}(Books)$

genre	AVG(price)
biography	35
fantasy	30
poetry	40

Semirings alone are insufficient

title	genre	price	
Book 1	biography	40	b_1
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Book 4	biography	30	b_4
Book 5	fantasy	10	b_5
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Let S be a commutative semiring. An S-*semimodule* $(M, +_M, 0_M, *)$ is a commutative monoid $(M, +_M, 0_M)$ such that for all $r, s \in S$ and $m, n \in M$:

$$r * (m +_M n) = r * m +_M r * n$$
$$(r +_S s) * m = r * m +_M s * m$$
$$(r \cdot_S s) * m = r * (s * m)$$
$$1_S * m = m$$
$$0_S * m = 0_M \quad \text{and} \quad r * 0_M = 0_M$$

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- Trade-offs between generality and practicality
- Limits of semiring-based provenance

Contributions

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- A structured accessible narrative

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- Proofs for propositions/theorems

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- A structured accessible narrative
- Proofs for propositions/theorems
- Reformulated concepts for clarity

Fundamental theorem

Let h be a semiring homomorphism. Then, for any $\mathcal{S}$-relational algebra query Q:

$$\begin{tikzcd}
\mathcal{S}_{db} \arrow[r, "h_{Rel}"] \arrow[d, "Q"] & \mathcal{S}'_{db} \arrow[d, "Q"] \\
\mathcal{S}_{rel} \arrow[r, "h_{Rel}"] & \mathcal{S}'_{rel}
\end{tikzcd}$$